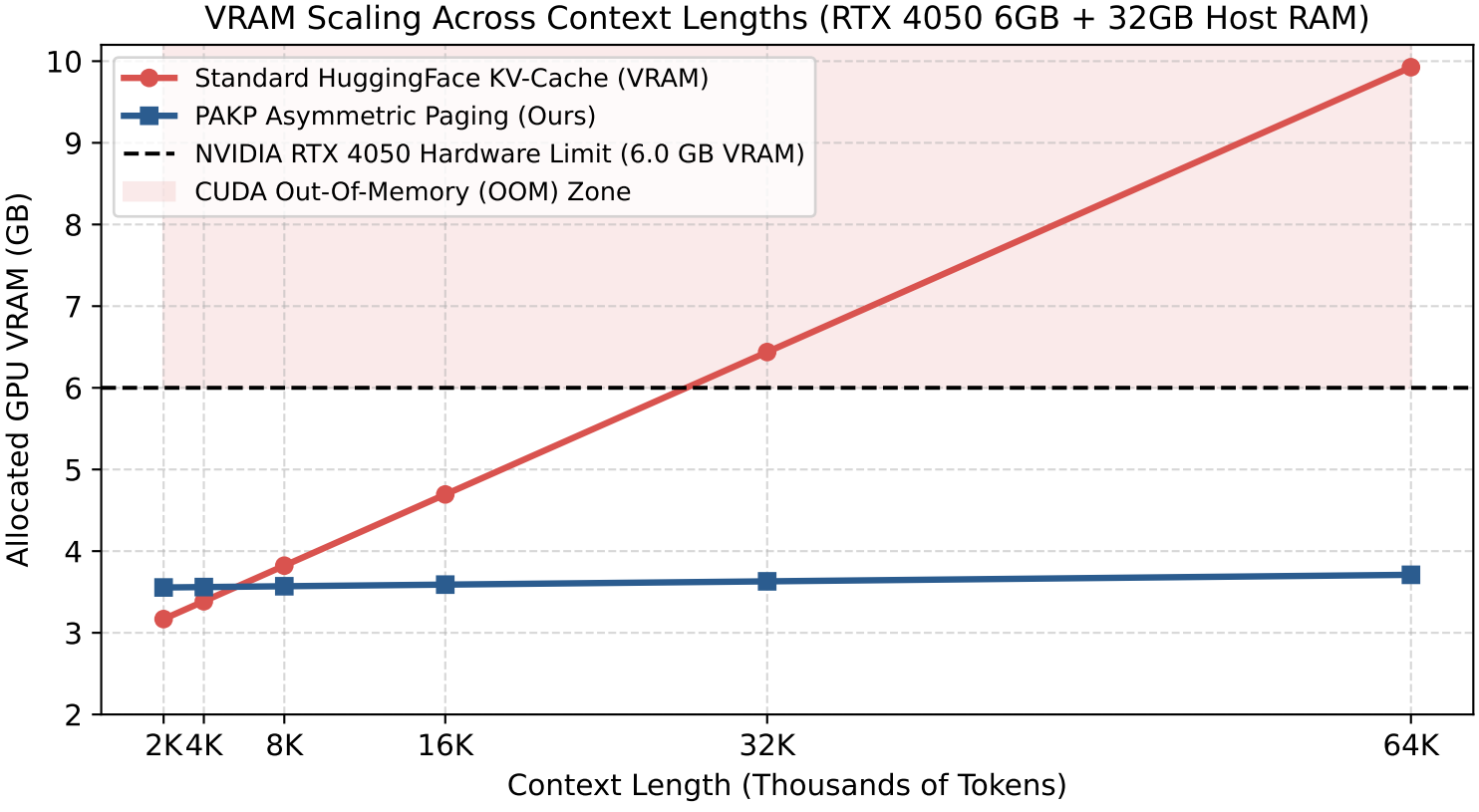
VRAM Scaling Across Context Lengths (RTX 4050 6GB + 32GB Host RAM)
Standard HuggingFace KV-Cache (VRAM)
PAKP Asymmetric Paging (Ours)
NVIDIA RTX 4050 Hardware Limit (6.0 GB VRAM)
CUDA Out-Of-Memory (OOM) Zone
Allocated GPU VRAM (GB)
10
9
8
7
6
5
4
3
2
2K
4K
8K
16K
32K
64K
Context Length (Thousands of Tokens)